

Team Locked Inc.

Texas A&M University
Department of Engineering Technology
& Industrial Distribution

ESET Senior Capstone, Spring 2026

Team Lead Member 2
Hardware Lead Software Lead

Project advisor: (TBD)
Sponsor: (pending TAMU Disability
Resources + OREC + Facilities)

QR

github.com/Locked-Inc/STAR
Open-source platform + compliance engine

We're seeking
pilot building partners.

Why this matters

The Americans with Disabilities Act has been federal law for 35 years. Industry estimates that roughly **73% of U.S. commercial buildings** still fail at least one ADA standard.

Only 32% of buyers ask at commercial inspection.

A Certified Access Specialist audit runs **\$800 to \$50,000+** depending on scope and takes **2 to 5 days** of manual measurement. For a campus like TAMU's - 5,200 acres, hundreds of buildings - full manual coverage is effectively impossible.

The bottleneck is measurement, not law.

STAR is the measurement layer. A distributed robot - Raspberry Pi 5 plus a custom RX72N real-time motor controller, SPI with HARQ and FEC, RPLiDAR C1 plus IMX219-83 stereo plus two IMUs - runs ROS2 Jazzy with slam_toolbox and Nav2 and our new ADA compliance engine on top. One check (ramp slope) is implemented end-to-end; six more are architected against the same sensor stack.

Same economic shift cybersecurity scanners brought to IT compliance a decade ago.

STAR

Spatial Topography Accessibility Robot

\$236,451

max DOJ penalty per ADA repeat violation

8,800

ADA Title III federal lawsuits in 2024

1 in 4

U.S. adults live with a disability

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The problem

\$236,451

max DOJ penalty per repeat violation (90 FR 29445, July 2025)

8,800

ADA Title III lawsuits in 2024 (Seyfarth Shaw, +7% YoY)

~70 million

U.S. adults with a disability, 28.7% of adults (CDC 2022 BRFSS)

Manual audit economics

\$800-\$50,000+ per building depending on scope.

2-5 days on-site with tape measure, smart level, clipboard.

Only 32% of buyers ask about ADA at commercial inspection.

TAMU: 5,200 acres, hundreds of buildings - full manual pass is infeasible.

Our solution: STAR

Distributed autonomous platform + ADA compliance engine.

Platform (demo-ready)

Raspberry Pi 5 (ROS2 Jazzy) + custom Renesas RX72N PCB (ThreadX).

SPI 10 Mbps with HARQ Chase Combining + FEC K=7 + nanopb + CRC-32.

slam_toolbox async + Nav2 + m-explore-ros2 frontier.

143 ROS2 tests passing, 0 failures.

Sensors

SLAMTEC RPLiDAR C1 - 360 deg, 10 Hz, 12 m, +/- 3 cm.

Waveshare IMX219-83 stereo - dual 8 MP, 60 mm baseline, dual CSI-2.

BNO055 9-DoF IMU on chassis + **ICM20948** 9-DoF IMU on camera.

4x HC-SR04 ultrasonic + DS18B20 + BMP280.

Compliance engine (7 ADA checks)

IMPLEMENTED: Ramp slope 405.2

STRETCH: Trip hazard 303, Path width 403.5

ARCHITECTED: Ramp width, Ramp landing, Door clear, Door threshold

Validation

Platform (complete)

143 / 143 ROS2 tests passing.

slam_toolbox /map at 0.5 Hz on RPi5; full TF chain.

Nav2 autonomous goals executed; frontier exploration verified.

SPI transport 1.6% utilization at 10 Mbps.

Ramp-slope compliance check

Ground truth: Wixey WR300 digital angle gauge (+/- 0.1 deg).

LiDAR plane normal and BNO055 pitch, agreement gate +/- 0.5 deg.

On-campus ramp; n per dress-rehearsal session; live validation_log.csv.

Impact at TAMU scale

5,200 acres, hundreds of buildings.

Manual audit = multi-million-dollar cost.

STAR = amortized hardware cost, re-runnable monthly.

Stakeholders: TAMU Disability Resources + OREC + Facilities.